

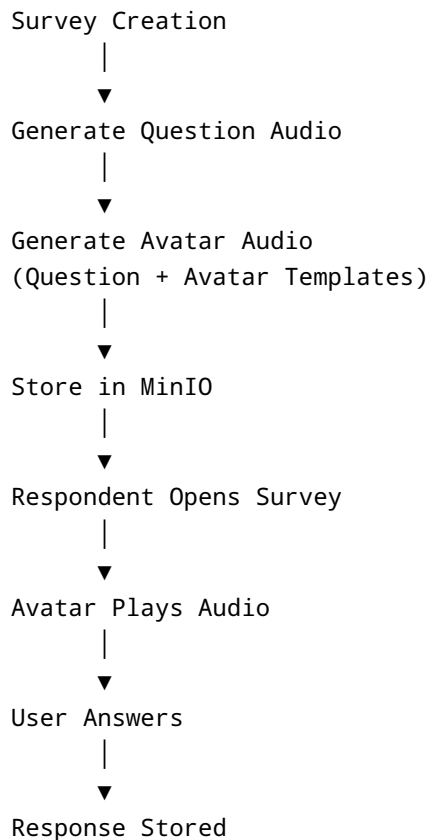
AI Avatar Survey System - End-to-End Implementation Plan

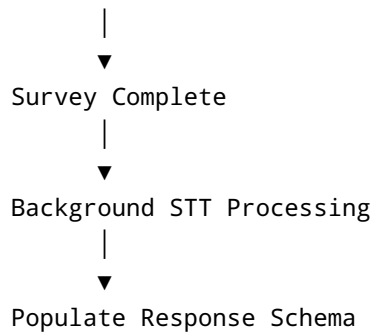
Objective

Create a web-based AI survey avatar that:

- Reuses existing survey audio infrastructure.
 - Uses a reusable 3D avatar.
 - Does not use conversational AI.
 - Does not use real-time STT.
 - Does not use real-time validation.
 - Records user responses.
 - Processes responses using Sarvam STT after survey completion.
 - Follows the same architecture philosophy as the IVR implementation.
-

High-Level Architecture





Phase 1: Audio Infrastructure

Existing

Current:

audioid123.mp3

Contains:

Question
Option 1
Option 2
Option 3

New Avatar Templates

Generate once per language.

English

Hello, I will help you complete this survey.

↓

avatar_intro_en.mp3

Please answer the question shown on your screen.

↓

avatar_outro_en.mp3

Hindi

avatar_intro_hi.mp3
avatar_outro_hi.mp3

Marathi

avatar_intro_mr.mp3
avatar_outro_mr.mp3

Avatar Audio Generation

During survey audio generation:

avatar_intro_en.mp3
+
audioid123.mp3
+
avatar_outro_en.mp3

↓

audioid123_avatar.mp3

Store in MinIO.

Naming Convention

```
uuid.mp3  
uuid_twilio.mp3  
uuid_avatar.mp3
```

Examples:

```
abc123.mp3  
abc123_twilio.mp3  
abc123_avatar.mp3
```

Phase 2: Survey Schema Changes

No MongoDB Changes Required

Keep:

```
{  
  "audio": {  
    "english": "abc123.mp3"  
  }  
}
```

Avatar filename derived dynamically.

```
function getAvatarAudio(filename) {  
  return filename.replace(".mp3", "_avatar.mp3");  
}
```

Phase 3: Frontend Avatar Layer

Technology Stack

```
React  
Three.js
```

```
React Three Fiber
Drei
```

Install:

```
npm install three
npm install @react-three/fiber
npm install @react-three/drei
```

Avatar Asset

Store:

```
public/
└─ interviewer.glb
```

Current downloaded GLB can be used.

Future:

```
professional_interviewer.glb
```

recommended.

Avatar Component

```
Avatar
├─ Load GLB
├─ Mouth Animation
├─ Blink Animation
├─ Head Movement
└─ Audio Synchronization
```

Phase 4: Audio Playback Engine

Flow

Question Loaded



Fetch

```
audioid123_avatar.mp3
```



Play Audio



Trigger Mouth Animation

Audio Analyser

```
Audio
↓
Web Audio API
↓
Amplitude
↓
Jaw Bone Rotation
```

Pseudo:

```
jaw.rotation.x = volume * 0.03;
```

Phase 5: Avatar Animations

Mandatory

Jaw Movement

While audio plays:

Open
Close
Open
Close

based on volume.

Blink

Every:

3-8 seconds

randomized.

Head Movement

Small idle motion:

left
right
center

using sine wave.

Optional

Future:

Rhubarb Lip Sync

for phoneme-based lip movement.

Not required for MVP.

Phase 6: MCQ Experience

Avatar Speaks

Gender

Option 1 Male

Option 2 Female

Option 3 Other

Modal Opens

Gender

- Male
- Female
- Other

[Continue]

Store Response

Current response schema unchanged.

Phase 7: Text Question Experience

Avatar Speaks

What improvements would you like to see?

Recording Panel

 Voice Response

Speak after the tone

• Recording...

00:12

[Stop Recording]

After Recording

✓ Recording Saved

Duration: 24 sec

[Play]

[Re-record]

[Continue]

Phase 8: Audio Recording

Browser Recording

Use:

```
MediaRecorder API
```

Generate:

```
response_abc123.webm
```

or

```
response_abc123.wav
```

Upload immediately.

Storage

MinIO

```
responses/  
  surveyId/  
    respondentId/  
      questionId.wav
```

Phase 9: Response Schema

Store:

```
{  
  "questionId": "q123",  
  "responseType": "audio",  
  "audioFile": "questionId.wav",  
  "transcription": null,  
}
```

```
"status": "pending"
}
```

Phase 10: Survey Completion

Respondent completes survey.

No STT performed.

Immediately show:

Thank you for completing the survey.

Phase 11: Background STT Pipeline

Trigger:

SurveyCompleted Event

↓

Fetch all audio files.

↓

Send to Sarvam STT.

↓

Receive transcript.

↓

Update response records.

Result

Before:

```
{
  "audioFile": "q1.wav",
  "transcription": null
}
```

After:

```
{
  "audioFile": "q1.wav",
  "transcription": "Roads should be repaired"
}
```

Phase 12: Accessibility

Support:

- Audio controls
- Replay question
- Replay avatar speech
- Large buttons
- Keyboard navigation
- Mobile responsive

Phase 13: Future Enhancements

Phase 2

Add:

```
Professional Avatar
```

instead of cartoon character.

Phase 3

Add:

Phoneme Lip Sync

using:

Rhubarb

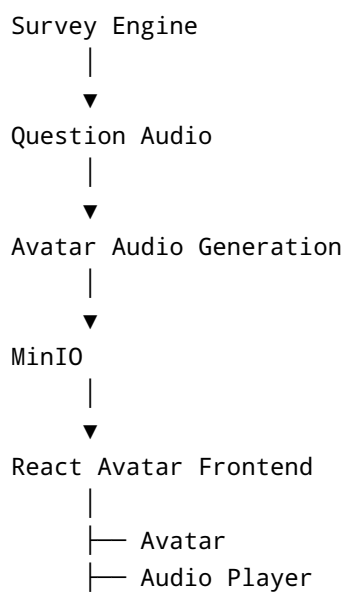
Phase 4

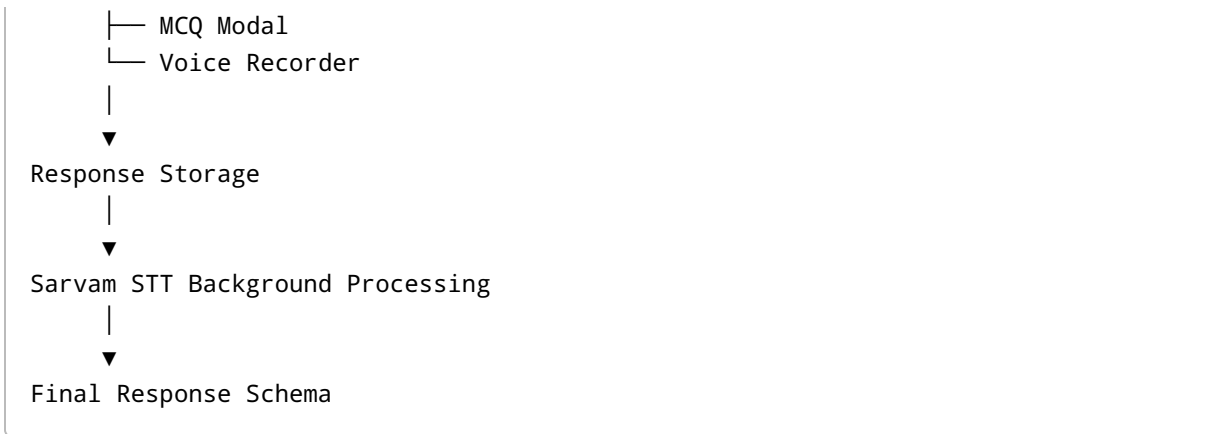
Add:

Emotion States

- Greeting
- Listening
- Thank You

Final Architecture





Core principle:

- Reuse existing survey audio pipeline.
- Reuse IVR stitching strategy.
- Use a single reusable 3D avatar.
- No conversational AI.
- No real-time STT.
- No per-question avatar generation costs.
- Scales similarly to the existing IVR architecture.